MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. The value of $\frac{\sin 30^\circ + \cos 60^\circ}{\tan 45^\circ}$ is: [1]

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) 2

Q2. If $\csc\theta = \frac{5}{3}$ and $\cot\phi = \frac{12}{5}$, where θ, ϕ are acute, then $\cos\theta \sin\phi$ is: [1]

(A) $\frac{4}{13}$ (B) $\frac{3}{13}$
(C) $\frac{4}{5}$ (D) $\frac{5}{13}$

Q3. If $2\sin^2\theta - \cos^2\theta = \frac{1}{2}$, then the value of acute angle θ is: [1]

(A) 30° (B) 60° (C) 45° (D) 90°

Q4. If $\sec\theta + \tan\theta = p$, then the value of $\sin\theta$ in terms of p is: [1]

(A) $\frac{p^2-1}{p^2+1}$ (B) $\frac{p^2+1}{p^2-1}$
(C) $\frac{2p}{p^2+1}$ (D) $\frac{2p}{p^2-1}$

Q5. If $\sin\theta - \cos\theta = \frac{1}{2}$, then the value of $\sin\theta\cos\theta$ is: [1]

(A) $\frac{3}{8}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Q6. **Assertion (A):** $\sec^2\theta + \csc^2\theta = \sec^2\theta\csc^2\theta$ for any acute angle θ.
Reason (R): $\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta} = \frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta\cos^2\theta} = \frac{1}{\sin^2\theta\cos^2\theta} = \sec^2\theta\csc^2\theta$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If $\sqrt{3}\sin\theta - \cos\theta = 0$ and $0^\circ < \theta < 90^\circ$, find the value of θ. [2]

Q8. Prove that: $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \sin A + \cos A$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, prove that $m^2 - n^2 = 4\sqrt{mn}$. [3]

Q10. Prove that: $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$. [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

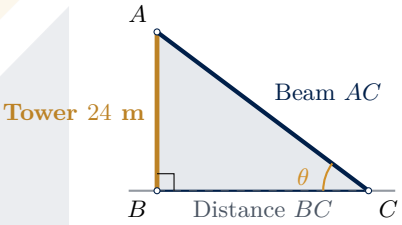
Q11. Prove that: $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x} = \frac{1}{\cos x} - \frac{1}{\sec x + \tan x}$. [5]

Q12. Prove that: $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{\sin^2 A - \cos^2 A} = \frac{2}{2\sin^2 A - 1}$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. High-Mast Stadium Floodlight Luminaire Projection

A sports stadium installs high-mast lighting towers to illuminate the soccer field. A mast AB of vertical height 24 m supports a bank of floodlights at top point A. The central axis of the floodlight beam hits the ground at point C, making an angle of elevation θ from the ground such that $\sin\theta = \frac{3}{5}$.



Based on the above geometric model, answer the following questions:

(i) Find the value of $\tan\theta$ and $\cos\theta$ for this beam angle. [1]

(ii) Calculate the ground distance BC from the base of the mast to the beam target point. [1]

(iii) Find the direct beam throw distance AC from the luminaires to the ground target, and verify the identity $1 + \cot^2\theta = \csc^2\theta$. [2]

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